Databricks Certified Data Engineer Associate- Practice Exam 1

Question 1Correct

Which of the following commands can a data engineer use to compact small data files of a Delta table into larger ones ?

- **PARTITION BY**
- **ZORDER BY**
- **COMPACT**
- **VACUUM**
- **OPTIMIZE (Your answer is correct)**

Overall explanation
Delta Lake can improve the speed of read queries from a table. One way to improve this speed is by compacting small files into larger ones. You trigger compaction by running the `OPTIMIZE` command

Reference: https://docs.databricks.com/sql/language-manual/delta-optimize.html

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Domain
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Question 2Correct

A data engineer is trying to use Delta time travel to rollback a table to a previous version, but the data engineer received an error that the data files are no longer present.

Which of the following commands was run on the table that caused deleting the data files?

- **VACUUM (Your answer is correct)**
- **OPTIMIZE**
- **ZORDER BY**
- **DEEP CLONE**
- **DELETE**

Overall explanation
Running the VACUUM command on a Delta table deletes the unused data files older than a specified data retention period. As a result, you lose the ability to time travel back to any version older than that retention threshold.

Reference: https://docs.databricks.com/sql/language-manual/delta-vacuum.html

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Question 3Correct

In Delta Lake tables, which of the following is the primary format for the data files?

- **Delta**
- **Parquet (Your answer is correct)**
- **JSON**
- **Hive-specific format**
- **Both, Parquet and JSON**

Overall explanation
Delta Lake builds upon standard data formats. Delta lake table gets stored on the storage in one or more data files in Parquet format, along with transaction logs in JSON format.

Reference: https://docs.databricks.com/delta/index.html

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Question 4Correct

Which of the following locations hosts the Databricks web application ?

- **Data plane**
- **Control plane (Your answer is correct)**
- **Databricks Filesystem**
- **Databricks-managed cluster**
- **Customer Cloud Account**

Overall explanation
According to the Databricks Lakehouse architecture, Databricks workspace is deployed in the control plane along with Databricks services like Databricks web application (UI), Cluster manager, workflow service, and notebooks.

Reference: https://docs.databricks.com/getting-started/overview.html

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Databricks Lakehouse Platform

Question 5Correct

In Databricks Repos, which of the following operations a data engineer can use to update the local version of a repo from its remote Git repository ?

- **Clone**
- **Commit**
- **Merge**
- **Push**
- **Pull (Your answer is correct)**

Overall explanation
The git Pull operation is used to fetch and download content from a remote repository and immediately update the local repository to match that content.

References:

- https://docs.databricks.com/repos/index.html
- https://github.com/git-guides/git-pull

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Question 6Correct

According to the Databricks Lakehouse architecture, which of the following is located in the customer's cloud account?

- **Databricks web application**
- **Notebooks**
- **Repos**
- **Cluster virtual machines (Your answer is correct)**
- **Workflows**

Overall explanation
When the customer sets up a Spark cluster, the cluster virtual machines are deployed in the data plane in the customer's cloud account.

Reference: https://docs.databricks.com/getting-started/overview.html

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Question 7Correct

Which of the following best describes Databricks Lakehouse?

- **Single, flexible, high-performance system that supports data, analytics, and machine learning workloads. (Your answer is correct)**
- **Reliable data management system with transactional guarantees for organization’s structured data.**
- **Platform that helps reduce the costs of storing organization’s open-format data files in the cloud.**
- **Platform for developing increasingly complex machine learning workloads using a simple, SQL-based solution.**
- **Platform that scales data lake workloads for organizations without investing on-premises hardware.**

Overall explanation
Databricks Lakehouse is a unified analytics platform that combines the best elements of data lakes and data warehouses. So, in the Lakehouse, you can work on data engineering, analytics, and AI, all in one platform.

Reference: https://www.databricks.com/glossary/data-lakehouse

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Databricks Lakehouse Platform

Question 8 **Correct**

If the default notebook language is SQL, which of the following options a data engineer can use to run a Python code in this SQL Notebook ?

- **They need first to import the python module in a cell**
- **This is not possible! They need to change the default language of the notebook to Python**
- **Databricks detects cells language automatically, so they can write Python syntax in any cell**
- **They can add `%language` magic command at the start of a cell to force language detection.**
- **They can add `%python` at the start of a cell. (Your answer is correct)**

Overall explanation
By default, cells use the default language of the notebook. You can override the default language in a cell by using the language magic command at the beginning of a cell. The supported magic commands are: `%python`, `%sql`, `%scala`, and `%r`.

Reference: https://docs.databricks.com/notebooks/notebooks-code.html

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Question 9 **Incorrect**

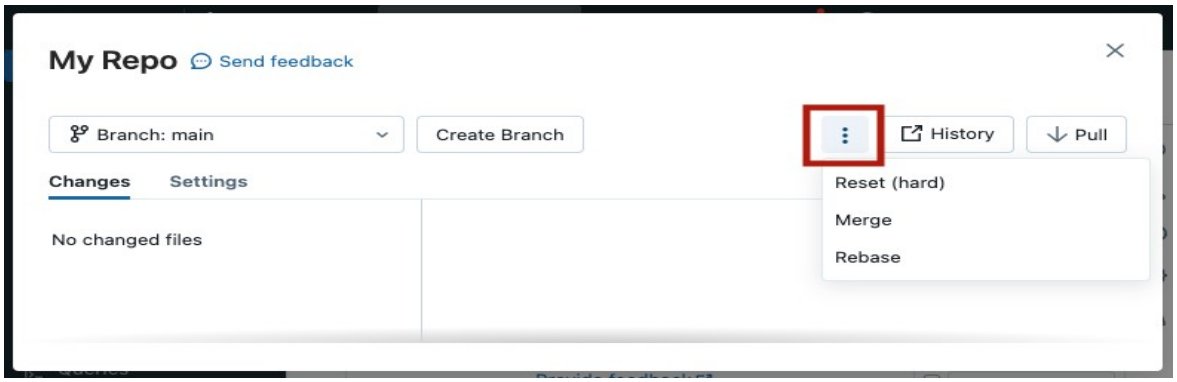
Which of the following tasks is not supported by Databricks Repos, and must be performed in your Git provider ?

- **Clone, push to, or pull from a remote Git repository.**
- **Create and manage branches for development work.**
- **Create notebooks, and edit notebooks and other files.**
- **Visually compare differences upon commit.**
- **Delete branches (Correct answer)**

Overall explanation
The following tasks are not supported by Databricks Repos, and must be performed in your Git provider:

- Create a pull request
- Delete branches
- Merge and rebase branches *

* **NOTE:** Recently, merge and rebase branches have become supported in Databricks Repos. However, this may still not be updated in the current exam version.



Reference: https://docs.databricks.com/repos/index.html

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Databricks Lakehouse Platform

Question 10 **Correct**

Which of the following statements is **Not** true about Delta Lake ?

- **Delta Lake provides ACID transaction guarantees**
- **Delta Lake provides scalable data and metadata handling**
- **Delta Lake provides audit history and time travel**
- **Delta Lake builds upon standard data formats: Parquet + XML (Correct answer)**
- **Delta Lake supports unified streaming and batch data processing**

Overall explanation
It is not true that Delta Lake builds upon XML format. It builds upon Parquet and JSON formats

Reference: https://docs.databricks.com/delta/index.html

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Databricks Lakehouse Platform

Question 11 **Correct**

How long is the default retention period of the VACUUM command ?

- **0 days**
- **7 days (Correct answer)**
- **30 days**
- **90 days**
- **365 days**

Overall explanation
By default, the retention threshold of the VACUUM command is 7 days. This means that VACUUM operation will prevent you from deleting files less than 7 days old, just to ensure that no long-running operations are still referencing any of the files to be deleted.

Reference: https://docs.databricks.com/sql/language-manual/delta-vacuum.html

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Question 12 **Correct**

The data engineering team has a Delta table called **employees** that contains the employees personal information including their gross salaries.

Which of the following code blocks will keep in the table only the employees having a salary greater than 3000 ?

- **DELETE FROM employees WHERE salary > 3000;**
- **SELECT CASE WHEN salary <= 3000 THEN DELETE ELSE UPDATE END FROM employees;**
- **UPDATE employees WHERE salary > 3000 WHEN MATCHED SELECT;**
- **UPDATE employees WHERE salary <= 3000 WHEN MATCHED DELETE;**
- **DELETE FROM employees WHERE salary <= 3000; (Correct answer)**

Overall explanation
In order to keep only the employees having a salary greater than 3000, we must delete the employees having salary less than or equal 3000. To do so, use the DELETE statement:

```
DELETE FROM table_name WHERE condition;
```

Reference: https://docs.databricks.com/sql/language-manual/delta-delete-from.html

Domain
ELT with Spark SQL and Python

Question 13 **Incorrect**

A data engineer wants to create a relational object by pulling data from two tables. The relational object must be used by other data engineers in other sessions on the same cluster only. In order to save on storage costs, the date engineer wants to avoid copying and storing physical data.

Which of the following relational objects should the data engineer create?

- **Temporary view**
- **External table**
- **Managed table**
- **Global Temporary view (Correct answer)**
- **View**

Overall explanation
In order to avoid copying and storing physical data, the data engineer must create a view object. A view in databricks is a virtual table that has no physical data. It’s just a saved SQL query against actual tables.

The view type should be Global Temporary view that can be accessed in other sessions on the same cluster. Global Temporary views are tied to a cluster temporary database called global_temp.

Reference: https://docs.databricks.com/sql/language-manual/sql-ref-syntax-ddl-create-view.html

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Domain
ELT with Spark SQL and Python

Question 14Correct

A data engineer has developed a code block to completely reprocess data based on the following if-condition in Python:

```
1. if process_mode = "init" and not is_table_exist:
2.     print("Start processing ...")
```

This if-condition is returning an invalid syntax error.

Which of the following changes should be made to the code block to fix this error ?

```
1. if process_mode = "init" & not is_table_exist:
2.     print("Start processing ...")
```

```
1. if process_mode = "init" and not is_table_exist = True:
2.     print("Start processing ...")
```

```
1. if process_mode = "init" and is_table_exist = False:
2.     print("Start processing ...")
```

```
1. if (process_mode = "init") and (not is_table_exist):
2.     print("Start processing ...")
```

```
1. if process_mode == "init" and not is_table_exist:
2.     print("Start processing ...")
```
(Correct answer)

Overall explanation
Python if statement looks like this in its simplest form:

```
1. if <expr>:
2.     <statement>
```

Python supports the usual logical conditions from mathematics:

- Equals: `a == b`
- Not Equals: `a != b`
- `<`, `<=`, `>`, `>=`

To combine conditional statements, you can use the following logical operators:

- `and`
- `or`

The negation operator in Python is: `not`

Reference: https://www.w3schools.com/python/python_conditions.asp

Domain
ELT with Spark SQL and Python

Question 15Correct

Fill in the below blank to successfully create a table in Databricks using data from an existing PostgreSQL database:

```
CREATE TABLE employees
  USING _____________
  OPTIONS (
    url "jdbc:postgresql:dbserver",
    dbtable "employees"
  )
```

- **org.apache.spark.sql.jdbc (Correct answer)**
- **postgresql**
- **DELTA**
- **dbserver**
- **cloudfiles**

Overall explanation
Using the JDBC library, Spark SQL can extract data from any existing relational database that supports JDBC. Examples include mysql, postgres, SQLite, and more.

Reference: https://learn.microsoft.com/en-us/azure/databricks/external-data/jdbc

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Domain
ELT with Spark SQL and Python

Question 16Correct

Which of the following commands can a data engineer use to create a new table along with a comment ?

```
CREATE TABLE payments
COMMENT "This table contains sensitive information"
AS SELECT * FROM bank_transactions
```
(Correct answer)

```
CREATE TABLE payments
COMMENT("This table contains sensitive information")
AS SELECT * FROM bank_transactions
```

```
CREATE TABLE payments
AS SELECT * FROM bank_transactions
COMMENT "This table contains sensitive information"
```

```
CREATE TABLE payments
AS SELECT * FROM bank_transactions
COMMENT("This table contains sensitive information")
```

```
COMMENT("This table contains sensitive information")
CREATE TABLE payments
AS SELECT * FROM bank_transactions
```

Overall explanation
The `CREATE TABLE` clause supports adding a descriptive comment for the table. This allows for easier discovery of table contents.

Syntax:

```
CREATE TABLE table_name
COMMENT "here is a comment"
AS query
```

Reference: https://docs.databricks.com/sql/language-manual/sql-ref-syntax-ddl-create-table-using.html

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Question 17 **Correct**

A junior data engineer usually uses `INSERT INTO` command to write data into a Delta table. A senior data engineer suggested using another command that avoids writing of duplicate records.

Which of the following commands is the one suggested by the senior data engineer ?

- **MERGE INTO (Correct answer)**
- **APPLY CHANGES INTO**
- **UPDATE**
- **COPY INTO**
- **INSERT OR OVERWRITE**

Overall explanation
MERGE INTO allows to merge a set of updates, insertions, and deletions based on a source table into a target Delta table. With MERGE INTO, you can avoid inserting the duplicate records when writing into Delta tables.

References:

- https://docs.databricks.com/sql/language-manual/delta-merge-into.html
- https://docs.databricks.com/delta/merge.html#data-deduplication-when-writing-into-delta-tables

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Question 18 **Incorrect**
A data engineer is designing a Delta Live Tables pipeline. The source system generates files containing changes captured in the source data. Each change event has metadata indicating whether the specified record was inserted, updated, or deleted. In addition to a timestamp column indicating the order in which the changes happened. The data engineer needs to update a target table based on these change events.

Which of the following commands can the data engineer use to best solve this problem?

- **MERGE INTO**
- **APPLY CHANGES INTO (Correct answer)**
- **UPDATE**
- **COPY INTO**
- **cloud_files**

Overall explanation
The events described in the question represent Change Data Capture (CDC) feed. CDC is logged at the source as events that contain both the data of the records along with metadata information:

1. Operation column indicating whether the specified record was inserted, updated, or deleted
2. Sequence column that is usually a timestamp indicating the order in which the changes happened

You can use the `APPLY CHANGES INTO` statement to use Delta Live Tables CDC functionality

Reference: https://docs.databricks.com/workflows/delta-live-tables/delta-live-tables-cdc.html

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ELT with Spark SQL and Python

Question 19Correct

In PySpark, which of the following commands can you use to query the Delta table **employees** created in Spark SQL?

- **pyspark.sql.read(SELECT * FROM employees)**
- **spark.sql("employees")**
- **spark.format(“sql”).read("employees")**
- **spark.table("employees") (Correct answer)**
- **Spark SQL tables can not be accessed from PySpark**

Overall explanation
`spark.table()` function returns the specified Spark SQL table as a PySpark DataFrame

Reference:

https://spark.apache.org/docs/2.4.0/api/python/_modules/pyspark/sql/session.html#SparkSession.table

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Domain
ELT with Spark SQL and Python

Question 20Correct
Which of the following code blocks can a data engineer use to create a user defined function (UDF) ?

- **CREATE FUNCTION plus_one(value INTEGER)**
  RETURN value +1
- **CREATE UDF plus_one(value INTEGER)**
  RETURNS INTEGER
  RETURN value +1;
- **CREATE UDF plus_one(value INTEGER)**
  RETURN value +1;

- **CREATE FUNCTION plus_one(value INTEGER)**
  RETURNS INTEGER
  RETURN value +1; (Correct answer)
- **CREATE FUNCTION plus_one(value INTEGER)**
  RETURNS INTEGER
  value +1;

Overall explanation
The correct syntax to create a UDF is:

```
1. CREATE [OR REPLACE] FUNCTION function_name ( [ parameter_name data_type [, ...] ] )
2. RETURNS data_type
3. RETURN { expression | query }
```

Reference: https://docs.databricks.com/udf/index.html

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ELT with Spark SQL and Python

Question 21Correct

When dropping a Delta table, which of the following explains why only the table's metadata will be deleted, while the data files will be kept in the storage ?

- **The table is deep cloned**
- **The table is external (Correct answer)**
- **The user running the command has no permission to delete the data files**
- **The table is managed**
- **Delta prevents deleting files less than retention threshold, just to ensure that no long-running operations are still referencing any of the files to be deleted**

Overall explanation
External (unmanaged) tables are tables whose data is stored in an external storage path by using a `LOCATION` clause.

When you run `DROP TABLE` on an external table, only the table's metadata is deleted, while the underlying data files are kept.

Reference: https://docs.databricks.com/lakehouse/data-objects.html#what-is-an-unmanaged-table

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Question 22Incorrect

Given the two tables **students_course_1** and **students_course_2**. Which of the following commands can a data engineer use to get all the students from the above two tables without duplicate records ?

```
1. SELECT * FROM students_course_1
2. CROSS JOIN
3. SELECT * FROM students_course_2
```

Correct answer

```
SELECT * FROM students_course_1
UNION
SELECT * FROM students_course_2
```

```
SELECT * FROM students_course_1
INTERSECT
SELECT * FROM students_course_2
```

```
SELECT * FROM students_course_1
OUTER JOIN
SELECT * FROM students_course_2
```

```
SELECT * FROM students_course_1
INNER JOIN
SELECT * FROM students_course_2
```

Overall explanation
With `UNION`, you can return the result of subquery1 plus the rows of subquery2

Syntax:

```
subquery1
UNION [ ALL | DISTINCT ]
subquery2
```

- If `ALL` is specified duplicate rows are preserved.
- If `DISTINCT` is specified the result does not contain any duplicate rows. This is the default.

Note that both subqueries must have the same number of columns and share a least common type for each respective column.

Reference: https://docs.databricks.com/sql/language-manual/sql-ref-syntax-qry-select-setops.html

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ELT with Spark SQL and Python

Question 23Correct
Given the following command:

```
CREATE DATABASE IF NOT EXISTS hr_db ;
```

In which of the following locations will the **hr_db** database be located?

- **dbfs:/user/hive/warehouse (Correct answer)**
- **dbfs:/user/hive/db_hr**
- **dbfs:/user/hive/databases/db_hr.db**
- **dbfs:/user/hive/databases**
- **dbfs:/user/hive**

Overall explanation
Since we are creating the database here without specifying a `LOCATION` clause, the database will be created in the default warehouse directory under dbfs:/user/hive/warehouse

Reference: https://docs.databricks.com/sql/language-manual/sql-ref-syntax-ddl-create-schema.html

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ELT with Spark SQL and Python

Question 24Incorrect

Given the following table **faculties**

faculty_id	faculty_name	students
F001	Faculty of Medicine	[{"student_id": "S000002501", "total_courses": "6"}, {"student_id": "S000004478", "total_courses": "2"}, {"student_id": "S000001572", "total_courses": "5"}, {"student_id": "S000003859", "total_courses": "1"}]
F002	Faculty of Economics	[{"student_id": "S000007415", "total_courses": "3"}, {"student_id": "S000001177", "total_courses": "4"}, {"student_id": "S000005631", "total_courses": "7"}, {"student_id": "S000001003", "total_courses": "6"}]
F003	Faculty of Engineering	[{"student_id": "S000007251", "total_courses": "2"}, {"student_id": "S000002415", "total_courses": "5"}]

Fill in the below blank to get the students enrolled in less than 3 courses from the array column **students**

```
1. SELECT
2.   faculty_id,
3.   students,
4.   ____________ AS few_courses_students
5. FROM faculties
```

- **TRANSFORM (students, total_courses < 3)**
- **TRANSFORM (students, i -> i.total_courses < 3)**
- **FILTER (students, total_courses < 3)**
- **FILTER (students, i -> i.total_courses < 3) (Correct answer)**
- **CASE WHEN students.total_courses < 3 THEN students**
  ELSE NULL
  END

Overall explanation
`filter(input_array, lamda_function)` is a higher order function that returns an output array from an input array by extracting elements for which the predicate of a lambda function holds.

Example:

Extracting odd numbers from an input array of integers:

```
SELECT filter(array(1, 2, 3, 4), i -> i % 2 == 1);
```

output: [1, 3]

References:

- https://docs.databricks.com/sql/language-manual/functions/filter.html
- https://docs.databricks.com/optimizations/higher-order-lambda-functions.html

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Domain
ELT with Spark SQL and Python

Question 25Correct

Given the following Structured Streaming query:

```
(spark.table("orders")
        .withColumn("total_after_tax", col("total")+col("tax"))
    .writeStream
        .option("checkpointLocation", checkpointPath)
        .outputMode("append")
         ._____________
        .table("new_orders")
)
```

Fill in the blank to make the query executes a micro-batch to process data every 2 minutes

- **trigger(once=”2 minutes”)**
- **trigger(processingTime=”2 minutes") (Correct answer)**
- **processingTime(”2 minutes")**
- **trigger(”2 minutes")**
- **trigger()**

Overall explanation
In Spark Structured Streaming, in order to process data in micro-batches at the user-specified intervals, you can use `processingTime` keyword. It allows to specify a time duration as a string.

Reference: https://docs.databricks.com/structured-streaming/triggers.html#configure-structured-streaming-trigger-intervals

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Domain
Incremental Data Processing

Question 26Correct

Which of the following is used by Auto Loader to load data incrementally?

- **DEEP CLONE**
- **Multi-hop architecture**
- **COPY INTO**
- **Spark Structured Streaming (Correct answer)**
- **Databricks SQL**

Overall explanation
Auto Loader is based on Spark Structured Streaming. It provides a Structured Streaming source called cloudFiles.

Reference: https://docs.databricks.com/ingestion/auto-loader/index.html

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Domain
Incremental Data Processing

Question 27Incorrect

Which of the following statements best describes Auto Loader ?

- **Auto loader allows applying Change Data Capture (CDC) feed to update tables based on changes captured in source data.**
- **Auto loader monitors a source location, in which files accumulate, to identify and ingest only new arriving files with each command run. While the files that have already been ingested in previous runs are skipped. (Correct answer)**
- **Auto loader allows cloning a source Delta table to a target destination at a specific version.**
- **Auto loader defines data quality expectations on the contents of a dataset, and reports the records that violate these expectations in metrics.**
- **Auto loader enables efficient insert, update, deletes, and rollback capabilities by adding a storage layer that provides better data reliability to data lakes.**

Overall explanation
Auto Loader incrementally and efficiently processes new data files as they arrive in cloud storage.

Reference: https://docs.databricks.com/ingestion/auto-loader/index.html

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Domain
Incremental Data Processing

Question 28Incorrect

A data engineer has defined the following data quality constraint in a Delta Live Tables pipeline:

```
CONSTRAINT valid_id EXPECT (id IS NOT NULL) _____________
```

Fill in the above blank so records violating this constraint will be added to the target table, and reported in metrics

- **ON VIOLATION ADD ROW**
- **ON VIOLATION FAIL UPDATE**
- **ON VIOLATION SUCCESS UPDATE**
- **ON VIOLATION NULL**
- **There is no need to add ON VIOLATION clause. By default, records violating the constraint will be kept, and reported as invalid in the event log (Correct answer)**

Overall explanation
By default, records that violate the expectation are added to the target dataset along with valid records, but violations will be reported in the event log
Reference:

https://learn.microsoft.com/en-us/azure/databricks/workflows/delta-live-tables/delta-live-tables-expectations

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Domain
Incremental Data Processing

Question 29Correct

The data engineer team has a DLT pipeline that updates all the tables once and then stops. The compute resources of the pipeline continue running to allow for quick testing.

Which of the following best describes the execution modes of this DLT pipeline ?

- **The DLT pipeline executes in Continuous Pipeline mode under Production mode.**
- **The DLT pipeline executes in Continuous Pipeline mode under Development mode.**
- **The DLT pipeline executes in Triggered Pipeline mode under Production mode.**
- **The DLT pipeline executes in Triggered Pipeline mode under Development mode. (Correct answer)**
- **More information is needed to determine the correct response**

Overall explanation
Triggered pipelines update each table with whatever data is currently available and then they shut down.

In Development mode, the Delta Live Tables system ease the development process by

- Reusing a cluster to avoid the overhead of restarts. The cluster runs for two hours when development mode is enabled.
- Disabling pipeline retries so you can immediately detect and fix errors.

Reference: https://docs.databricks.com/workflows/delta-live-tables/delta-live-tables-concepts.html

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Domain
Incremental Data Processing

Question 30Correct

Which of the following will utilize Gold tables as their source?

- **Silver tables**
- **Auto loader**
- **Bronze tables**
- **Dashboards (Correct answer)**
- **Streaming jobs**

Overall explanation
Gold tables provide business level aggregates often used for reporting and dashboarding, or even for Machine learning

Reference: https://www.databricks.com/glossary/medallion-architecture

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Domain
Incremental Data Processing

Question 31Correct

Which of the following code blocks can a data engineer use to query the existing streaming table **events** ?

- **spark.readStream("events")**
- **spark.read.table("events")**
- **spark.readStream.table("events") (Correct answer)**
- **spark.readStream().table("events")**
- **spark.stream.read("events")**

Overall explanation
Delta Lake is deeply integrated with Spark Structured Streaming. You can load tables as a stream using:

```
spark.readStream.table(table_name)
```

Reference: https://docs.databricks.com/structured-streaming/delta-lake.html

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Domain
Incremental Data Processing

Question 32Correct

In multi-hop architecture, which of the following statements best describes the Bronze layer ?

- **It maintains data that powers analytics, machine learning, and production applications**
- **It maintains raw data ingested from various sources (Correct answer)**
- **It represents a filtered, cleaned, and enriched version of data**
- **It provides business-level aggregated version of data**
- **It provides a more refined view of the data.**

Overall explanation
Bronze tables contain data in its rawest format ingested from various sources (e.g., JSON files, Operational Databaes, Kakfa stream, ...)

Reference:https://www.databricks.com/glossary/medallion-architecture

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Domain
Incremental Data Processing

Question 33Correct

Given the following Structured Streaming query

```
1. (spark.readStream
2.         .format("cloudFiles")
3.         .option("cloudFiles.format", "json")
4.         .load(ordersLocation)
5.      .writeStream
6.         .option("checkpointLocation", checkpointPath)
7.         .table("uncleanedOrders")
```

```
8. )
```

Which of the following best describe the purpose of this query in a multi-hop architecture?

- **The query is performing raw data ingestion into a Bronze table (Correct answer)**
- **The query is performing a hop from a Bronze table to a Silver table**
- **The query is performing a hop from Silver table to a Gold table**
- **The query is performing data transfer from a Gold table into a production application**
- **This query is performing data quality controls prior to Silver layer**

Overall explanation
The query here is using Autoloader (**cloudFiles**) to load raw **json** data
from **ordersLocation** into the Bronze table **uncleanedOrders**

References:

- https://www.databricks.com/glossary/medallion-architecture
- https://docs.databricks.com/ingestion/auto-loader/index.html

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Domain
Incremental Data Processing

Question 34Incorrect

A data engineer has the following query in a Delta Live Tables pipeline:

```
1. CREATE LIVE TABLE aggregated_sales
2. AS
3.   SELECT store_id, sum(total)
4.   FROM cleaned_sales
5.   GROUP BY store_id
```

The pipeline is failing to start due to an error in this query

Which of the following changes should be made to this query to successfully start the DLT pipeline ?

```
1. CREATE STREAMING TABLE aggregated_sales
2. AS
3.   SELECT store_id, sum(total)
4.   FROM LIVE.cleaned_sales
5.   GROUP BY store_id
```

```
1. CREATE TABLE aggregated_sales
2. AS
3.   SELECT store_id, sum(total)
4.   FROM LIVE.cleaned_sales
5.   GROUP BY store_id
```

Correct answer

```
1. CREATE LIVE TABLE aggregated_sales
2. AS
3.   SELECT store_id, sum(total)
4.   FROM LIVE.cleaned_sales
5.   GROUP BY store_id
```

```
CREATE STREAMING LIVE TABLE aggregated_sales
AS
  SELECT store_id, sum(total)
  FROM cleaned_sales
  GROUP BY store_id
```

```
CREATE STREAMING LIVE TABLE aggregated_sales
AS
  SELECT store_id, sum(total)
  FROM STREAM(cleaned_sales)
  GROUP BY store_id
```

Overall explanation
In DLT pipelines, we use the `CREATE LIVE TABLE` syntax to create a table with SQL. To query another live table, prepend the `LIVE.` keyword to the table name.

CREATE LIVE TABLE aggregated_sales
AS
SELECT store_id, sum(total)
FROM **LIVE.**cleaned_sales
GROUP BY store_id

Reference: https://docs.databricks.com/workflows/delta-live-tables/delta-live-tables-sql-ref.html

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Domain
Incremental Data Processing
Question 35Incorrect
A data engineer has defined the following data quality constraint in a Delta Live Tables pipeline:

```
CONSTRAINT valid_id EXPECT (id IS NOT NULL) _____________
```

Fill in the above blank so records violating this constraint will be dropped, and reported in metrics

- **ON VIOLATION DROP ROW (Correct answer)**
- **ON VIOLATION FAIL UPDATE**
- **ON VIOLATION DELETE ROW**
- **ON VIOLATION DISCARD ROW**
- **There is no need to add ON VIOLATION clause. By default, records violating the constraint will be discarded, and reported as invalid in the event log**

Overall explanation
With `ON VIOLATION DROP ROW`, records that violate the expectation are dropped, and violations are reported in the event log

Reference:

https://learn.microsoft.com/en-us/azure/databricks/workflows/delta-live-tables/delta-live-tables-expectations

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Domain

Incremental Data Processing

Question 36Correct

Which of the following compute resources is available in Databricks SQL ?

- **Single-node clusters**
- **Multi-nodes clusters**
- **On-premises clusters**
- **SQL warehouses (Correct answer)**
- **SQL engines**

Overall explanation
Compute resources are infrastructure resources that provide processing capabilities in the cloud. A SQL warehouse is a compute resource that lets you run SQL commands on data objects within Databricks SQL.

Reference: https://docs.databricks.com/sql/admin/sql-endpoints.html

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Domain
Production Pipelines

Question 37Incorrect

Which of the following is the benefit of using the Auto Stop feature of Databricks SQL warehouses ?

- **Improves the performance of the warehouse by automatically stopping ideal services**
- **Minimizes the total running time of the warehouse (Correct answer)**
- **Provides higher security by automatically stopping unused ports of the warehouse**
- **Increases the availability of the warehouse by automatically stopping long-running SQL queries**
- **Databricks SQL does not have Auto Stop feature**

Overall explanation
The Auto Stop feature stops the warehouse if it’s idle for a specified number of minutes.

Reference: https://docs.databricks.com/sql/admin/sql-endpoints.html

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Domain
Production Pipelines

Question 38Correct

Which of the following alert destinations is **Not** supported in Databricks SQL ?

- **Slack**
- **Webhook**
- **SMS (Correct answer)**
- **Microsoft Teams**

- **Email**

Overall explanation
SMS is not supported as an alert destination in Databricks SQL . While, email, webhook, Slack, and Microsoft Teams are supported alert destinations in Databricks SQL.

Reference: https://docs.databricks.com/sql/admin/alert-destinations.html

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Domain
Production Pipelines

Question 39Correct

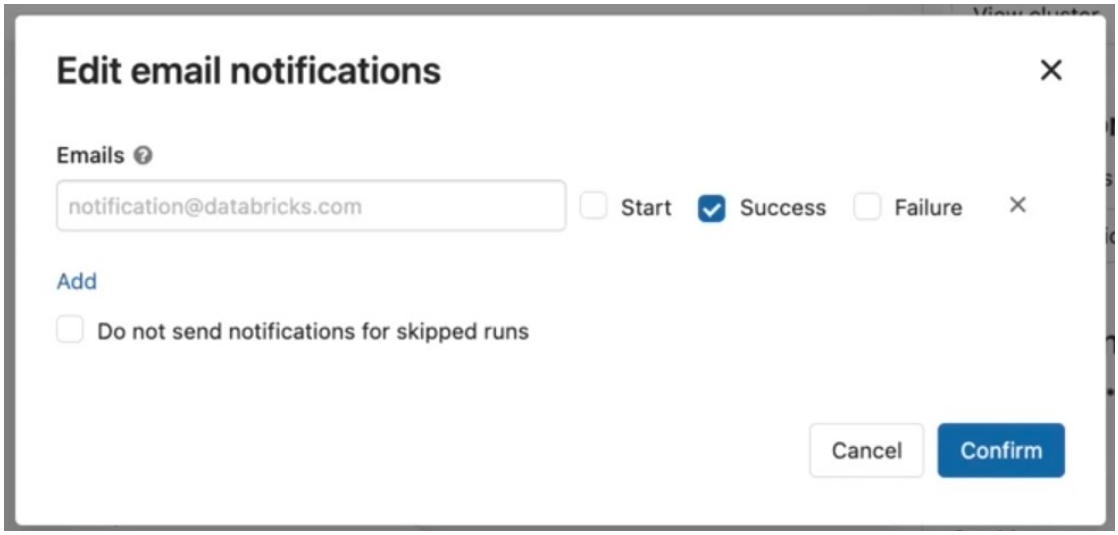
A data engineering team has a long-running multi-tasks Job. The team members need to be notified when the run of this job completes.

Which of the following approaches can be used to send emails to the team members when the job completes ?

- **They can use Job API to programmatically send emails according to each task status**
- **They can configure email notifications settings in the job page (Correct answer)**
- **There is no way to notify users when the job completes**
- **Only Job owner can be configured to be notified when the job completes**
- **They can configure email notifications settings per notebook in the task page**

Overall explanation

Databricks Jobs supports email notifications to be notified in the case of job start, success, or failure. Simply, click **Edit email notifications** from the details panel in the Job page. From there, you can add one or more email addresses.



Reference: https://docs.databricks.com/workflows/jobs/jobs.html#alerts-job

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Domain

Production Pipelines

Question 40Correct

A data engineer wants to increase the cluster size of an existing Databricks SQL warehouse.

Which of the following is the benefit of increasing the cluster size of Databricks SQL warehouses ?

- **Improves the latency of the queries execution (Correct answer)**
- **Speeds up the start up time of the SQL warehouse**
- **Reduces cost since large clusters use Spot instances**
- **The cluster size of SQL warehouses is not configurable. Instead, they can increase the number of clusters**
- **The cluster size can not be changed for existing SQL warehouses. Instead, they can enable the auto-scaling option.**

Overall explanation
Cluster Size represents the number of cluster workers and size of compute resources available to run your queries and dashboards. To reduce query latency, you can increase the cluster size.

Reference: https://docs.databricks.com/sql/admin/sql-endpoints.html#cluster-size-1

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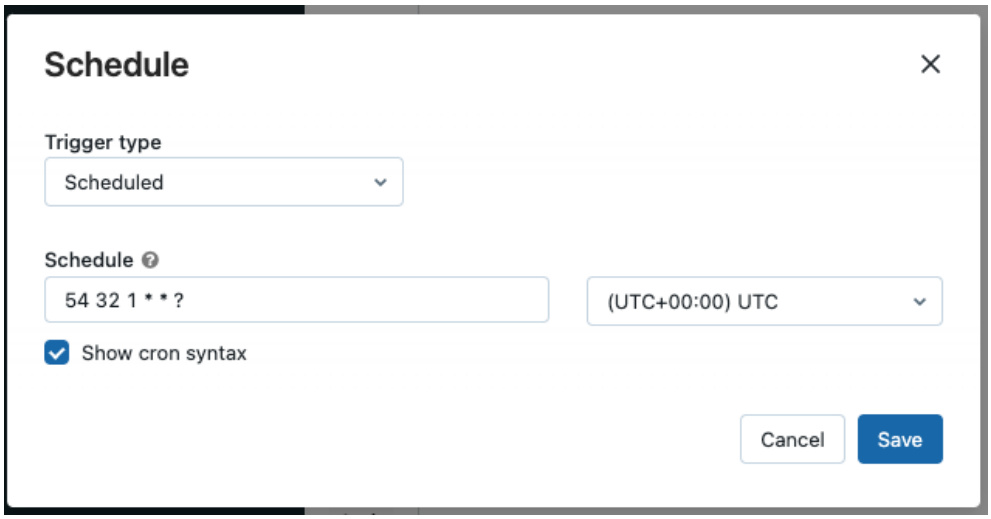
Domain
Production Pipelines

Question 41Correct

Which of the following describes Cron syntax in Databricks Jobs ?

- **It’s an expression to represent the maximum concurrent runs of a job**
- **It’s an expression to represent complex job schedule that can be defined programmatically (Correct answer)**
- **It’s an expression to represent the retry policy of a job**
- **It’s an expression to describe the email notification events (start, success, failure)**
- **It’s an expression to represent the run timeout of a job**

Overall explanation
To define a schedule for a Databricks job, you can either interactively specify the period and starting time, or write a Cron Syntax expression. The Cron Syntax allows to represent complex job schedule that can be defined programmatically



Reference: https://docs.databricks.com/workflows/jobs/jobs.html#schedule-a-job

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Domain
Production Pipelines

Question 42 **Incorrect**

The data engineer team has a DLT pipeline that updates all the tables at defined intervals until manually stopped. The compute resources terminate when the pipeline is stopped.

Which of the following best describes the execution modes of this DLT pipeline ?

- **The DLT pipeline executes in Continuous Pipeline mode under Production mode. (Correct answer)**
- **The DLT pipeline executes in Continuous Pipeline mode under Development mode.**
- **The DLT pipeline executes in Triggered Pipeline mode under Production mode.**
- **The DLT pipeline executes in Triggered Pipeline mode under Development mode.**
- **More information is needed to determine the correct response**

Overall explanation
Continuous pipelines update tables continuously as input data changes. Once an update is started, it continues to run until the pipeline is shut down.

In Production mode, the Delta Live Tables system:

- Terminates the cluster immediately when the pipeline is stopped.
- Restarts the cluster for recoverable errors (e.g., memory leak or stale credentials).
- Retries execution in case of specific errors (e.g., a failure to start a cluster)

Reference:https://docs.databricks.com/workflows/delta-live-tables/delta-live-tables-concepts.html

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Domain
Production Pipelines

Question 43Correct

Which part of the Databricks Platform can a data engineer use to grant permissions on tables to users ?

- **Data Studio**
- **Cluster event log**
- **Workflows**
- **DBFS**
- **Data Explorer (Correct answer)**

Overall explanation
Data Explorer in Databricks SQL allows you to manage data object permissions. This includes granting privileges on tables and databases to users or groups of users.

Reference: https://docs.databricks.com/security/access-control/data-acl.html#data-explorer

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Domain
Data Governance

Question 44Correct
Which of the following commands can a data engineer use to grant full permissions to the HR team on the table **employees** ?

- **GRANT FULL PRIVILEGES ON TABLE employees TO hr_team**
- **GRANT FULL PRIVILEGES ON TABLE hr_team TO employees**
- **GRANT ALL PRIVILEGES ON TABLE employees TO hr_team (Correct answer)**
- **GRANT ALL PRIVILEGES ON TABLE hr_team TO employees**
- **GRANT SELECT, MODIFY, CREATE, READ_METADATA ON TABLE employees TO hr_team**

Overall explanation
`ALL PRIVILEGES` is used to grant full permissions on an object to a user or group of users. It is translated into all the below privileges:

- `SELECT`
- `CREATE`
- `MODIFY`
- `USAGE`
- `READ_METADATA`

Reference: https://docs.databricks.com/security/access-control/table-acls/object-privileges.html#privileges

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Domain
Data Governance

Question 45Correct

A data engineer uses the following SQL query:

```
GRANT MODIFY ON TABLE employees TO hr_team
```

Which of the following describes the ability given by the `MODIFY` privilege ?

- **It gives the ability to add data from the table**
- **It gives the ability to delete data from the table**
- **It gives the ability to modify data in the table**
- **All the above abilities are given by the `MODIFY` privilege (Correct answer)**
- **None of these options correctly describe the ability given by the `MODIFY` privilege**

Overall explanation
The `MODIFY` privilege gives the ability to add, delete, and modify data to or from an object.

Reference: https://docs.databricks.com/security/access-control/table-acls/object-privileges.html#privileges

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Domain
Data Governance